Rate dependency of incremental capacity analysis (dQ/dV) as a diagnostic tool for lithium-ion batteries

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Abstract

Incremental capacity analysis (ICA) is a widely used method of characterising state of health (SOH) in secondary batteries through the identification of peaks that correspond to active material phase transformations. For reliable ICA, cells are cycled under low constant currents to minimise resistance and diffusion effects, making deployment into applications such as electric vehicle charging unfeasible.

In this work, the influence of charge/discharge rate on ICA is quantitively analysed through peak detection algorithms on two lithium-ion cells with different positive electrodes. Based on these results, a new robust method for faster ICA is introduced which corrects peak shift through SOC dependant resistance measurements using current interrupt. The new technique is evaluated through degradation tests on a $Li(NiCoAl)O_2$/graphite cell. Results demonstrate that ICA during a 6-hour (C/6) charge represents an ideal compromise between diagnostic accuracy and realistic application charge times. ICA at C/6 can predict peak location within 0.59% of a 48-hour charge (C/48) using resistance correction, compared to 1.90% without correction. Under aging, the C/6 charge was able to correctly identify the trend of each peak compared to C/24 charge and maintain peak location to within 2.0%.

At rates higher than C/6, the number of identifiable peaks in the ICA reduce, most noticeably in aged cells. After 200 cycles, only one identifiable peak was seen at 1C charge compared to four at C/6 and five at C/24.

Keywords: Differential capacity analysis; Incremental capacity analysis; battery; diagnostics; state of health

1. Introduction

Energy storage applications ranging from consumer electronics to electric vehicles and grid energy storage share a common requirement for high performance, low cost, durable and reliable lithium-ion batteries. With current technology, the lifetime of the battery is often less than the desired product lifetime, meaning the battery may need to be replaced or the product capability is reduced. One way to improve battery lifetime and device safety is through understanding how usage effects state of health (SOH) and managing its operation to reduce degradation rates. Due to the complex degradation mechanisms in batteries and varied duty cycles, SOH is not a directly measurable quantity but is inferred from current and historic measurements taken from the cell during its lifetime [1]. An ideal on-board SOH estimator should:

- Not negatively impact consumer use
- Have a low computational effort
- Utilise existing sensors
- Isolate degradation mechanisms

Many different methods for SOH estimation have been explored in the literature, reviewed by [2], including coulomb counting, current interrupt, electrochemical impedance spectroscopy (EIS), incremental capacity analysis (ICA), differential voltage analysis (DVA) and empirical models. However, there is still significant scope for further development in this area, especially in the identification of degradation modes.

ICA, otherwise known as differential capacity analysis (DCA), is a widely used technique to elucidate understanding of mechanisms occurring in the electrodes of a battery under constant current charge or discharge. Defined as the rate of change of charge with voltage (dQ/dV), peaks in the ICA, caused by plateaus in the voltage profile, correspond to phase transitions in the active electrode material due to intercalation [3]. Through monitoring the location, magnitude, width and area of these peaks, it is possible to study how the cell electrochemistry changes over the lifetime of a battery and to estimate its SOH [4].

The use of ICA was first introduced in the late 1960s [5] and has since evolved into a standard tool for non-destructive performance analysis of lithium-ion batteries. More recently, attempts have been made to directly correlate the change in ICA peaks to degradation mechanisms for improved SOH management. Dubarry [4], [6]–[18] has conducted significant work in this area, including the development of a mechanistic model which has been used to correlate experimental ICA to degradation mechanisms such as loss of active material (LAM), loss of lithium inventory (LLI) and ohmic resistance increase (ORI) [4] .

Differentiating experimental data amplifies noise present in the measurement signals, masking small changes in the ICA profile between consecutive cycles, leading many researchers to smooth data before processing [2]. An alternative approach taken by Smith et al. [19] was to develop high precision battery cyclers, enabling both accurate coulombic efficiency and determination of the change in ICA between two consecutive cycles, referred to as delta differential capacity analysis [20].

Weng et al. [21] extended the use of ICA from the cell to the pack level, addressing the problem of characterising multiple cells connected in series and parallel, and battery pack modules with aged cells and unequal current distribution. A methodology for tracking peaks in the ICA curve of battery modules was presented which utilised a support vector regression algorithm derived in previous work [22].

A key advantage of ICA over other in-situ characterisation techniques, such as electrochemical impedance spectroscopy (EIS), is that no additional equipment is required other than voltage sensing and current control, making it suitable for widespread use in many applications [23]. However, a potential drawback of using ICA in applications is that the cell needs to be cycled at a low current to minimise ohmic resistive effects altering the voltage at which peaks occur. The most widely used rates in the literature for ICA are C/24 [19], [20] and C/25 [10], [14], representing a compromise between quasi-steady-state open circuit voltage across the whole state of charge (SOC) and experiment time [7]. This can present challenges in high power transient applications like electric vehicles where usage patterns are user controlled and low rate constant current operation over a long period would negatively impact the user experience.

Several authors have used higher C-rates for ICA [7], [8], [24]–[27] predominately during aging studies. Merla et al. [25] compared ICA to differential thermal voltammetry for a $Ni_xMn_yCo_z$/graphite cell under 2C constant current charge, identifying two distinct peaks in the ICA. The temperature response was then used to identify an aged cell in a 4-cell pack. Samad et al. [28] compared force-based ICA to voltage-based ICA over cycles up to 1C, correlating the pressure change in cells to volume changes in the active material due to intercalation. Dubarry et al. [7] used ICA to study the degradation of $LiFePO_4$/graphite cells. ICA was presented at rates ranging from C/25 to C/2 and polarisation compensation applied to the voltage using resistance calculated from the linear section of the Tafel equation. ICA peak magnitude reduced, and peak width widened as current increased, potentially losing information for SOH analysis. Whilst this work introduced the concept of polarisation compensation at higher currents, comparing their relative accuracy and ability to analyse SOH was beyond the scope of work. Furthermore, a methodology for obtaining a polarisation corrected ICA from a single charge/discharge event was not presented. More recently Riviere et al. [27] investigated the change in a single ICA peak area with C-rate on a $LiFePO_4$/Graphite cell to estimate cell capacity through aging. To avoid changes in ICA peak area with C-rate, a 30-minute pause was implemented into the charging procedure at 50% SoC to allow the cell to reach thermodynamic equilibrium. Scindler et al. [29] conducted a combined experimental and model based study into how C-rate and temperature kinetic effects can be replicated in ICA using the Alawa model. They concluded that C-rate and temperature could be compensated through fitting of the ORI and rate degradation factor (RDF) degradation parameters.

DVA (dV/dQ) is the inverse of ICA, with peaks in the DVA corresponding to phase equilibria [30]. DVA is widely used for degradation studies in lithium-ion cells, and like ICA requires low constant current rates for accurate results. Dahn et al. [31] used DVA at C/20 and C/24 to estimate electrode slippage under aging by combining full cell and half-cell data. Bloom et al. [30] used full and half-cell DVA at C/25, to diagnose degradation mechanisms in calendar aged 18650 cells. Higher C-rate DVA was demonstrated by Lewerenz et al. [32], who used DVA at C/4 to characterise calendar aging in $LiFePO_4$/graphite cells with a focus on inhomogeneous distribution of lithium in the anode, which was correlated to DVA peak height characteristics. This work was then extended to calendar and cycling aging of $LiNiMnCoO_2$/graphite pouch cells by [33] and [34] respectively.

Despite the widespread use of ICA, there are very few quantitative studies on how charge/discharge rates in fresh and aged cells influence its accuracy as a tool for predicting battery SOH. The scope of this paper is to investigate the influence of C-rate on ICA and establish the maximum C-rate at which useful information can be extracted for degradation diagnostics. Two lithium-ion batteries utilising different positive electrode materials are quantitively studied, and a new method of increasing the rate of ICA whilst minimising peak voltage shift based on polarisation compensation is presented based on current interrupt. Finally, the ability of the new method to estimate SOH will be evaluated through degradation tests.

2. Experimental setup

Two different commercial 18650 cells were used to study the influence of positive electrode material on ICA at different currents. Cell A: 3350 mAh $LiNiCoAlO_2$/graphite cell (NCR18650b, Panasonic) and cell B: 1200 mAh $LiFePO_4$/graphite cell (818-3005, RS components). It is believed that both cells use a similar $LiPF_6$ /EC+DMC electrolyte, although the exact composition of each cell is not available in the public domain.

	Nominal capacity (mAh)	Nominal voltage (V)	Positive electrode	Negative electrode
Cell A	3350	3.7	$LiNiCoAlO_2$	Graphite
Cell B	1200	3.2	$LiFePO_4$	Graphite

Table 1- Cell specification

Cells were cycled using a Keithley 2461 four-quadrant source measure unit (SMU) in a four-wire setup and current and voltage sampled at 4 Hz with C-rates ranging from C/72 to 1C. As identified by [19], high accuracy voltage measurement and current control with low noise are important for reliable ICA. The current source error ranged from ±0.014% to ±0.620% depending on the C-rate used, increasing for higher currents. The maximum voltage measurement error was ±0.065%.

Cells were placed in a high current aluminium fixture and placed in a thermal chamber set to 25±1.0 °C for all tests. Temperature was recorded using three type-K thermocouples, two to record the cell surface temperature and one for the chamber temperature surrounding the cell. Temperature data was recorded using a PICO TC-08 datalogger at a 0.5 Hz sample rate. Electrochemical impedance spectroscopy tests were conducted using a Solartron 1280c.

A constant current – constant voltage (CC-CV) charging procedure was used with a 100 mA cut-off threshold. Cell A was charged to 4.2 V and cell B to 3.65 V. Cells were discharged at constant current until the manufacturer's minimum voltage (cell A 2.5 V, cell B 2.0 V). A rest period of 30 minutes was undertaken after each charge and discharge. Prior to testing, all fresh cells were conditioned using two initial charge and discharge cycles at 500 mA following the above procedure.

For the aging test, a fresh cell A was first conditioned, cycled at C/24 and C/6, then underwent cycling at 1C with CC-CV charge and constant current discharge. The C/24 and C/6 characterisation cycles were repeated after every 100 cycles. All C-rates are based on the manufactures quoted nominal capacity. C/6 was selected based on the results of the C-rate study (section 4.1) and because it represents a realistic time that an electric vehicle may be plugged in for continuous standard charging, either at a workplace or at home (overnight). A 30-minute rest period is included at the end of every charge and discharge event. This rest period is sufficient to allow for thermal equilibrium and voltage relaxation, but does not provide sufficient time for lithium diffusion into and out of the negative electrode overhang as discussed by [35], [36].

3. Methodology and data processing

3.1. Incremental capacity analysis

ICA was obtained from the voltage and current time histories during constant current charge and discharge. Each charge and discharge was split into 200 sections of equal time and the voltage and capacity change across each time period calculated. The incremental capacity change for each section was then calculated using equation 1, no smoothing of data was performed to obtain the ICA plots.

$$\left(\frac{dQ}{dV}\right)_n = \frac{Q_n - Q_{n+1}}{V_n - V_{n+1}} \tag{1}$$

Peak identification was performed on the ICA profiles using the 'findpeaks' function in MATLAB [37], with the minimum distance between adjacent peaks of four data points and minimum peak width of two data points (out of the 200 data points per ICA). Peak width is defined as the width of a peak at the vertical distance equal to half the peak prominence, where the prominence of a peak is its measured height relative to other surrounding peaks [37]. Peak area is calculated using trapezoidal integration of the ICA curve across the width of the peak, centred at the peak location. To reduce the number of false peaks identified through measurement noise, data was smoothed using a moving average of 5 points prior to identifying peaks.

In this work, peak location is identified as the voltage at which the local maximum absolute dQ/dV value occurs. Utilising the absolute value ensures that peaks locations correspond during charge (positive current) and discharge (negative current) ICA curves. Deviations between the charge and discharge peak locations will be influenced by polarisation and kinetic effects which will increase with C-rate.

3.2. Peak shift compensation

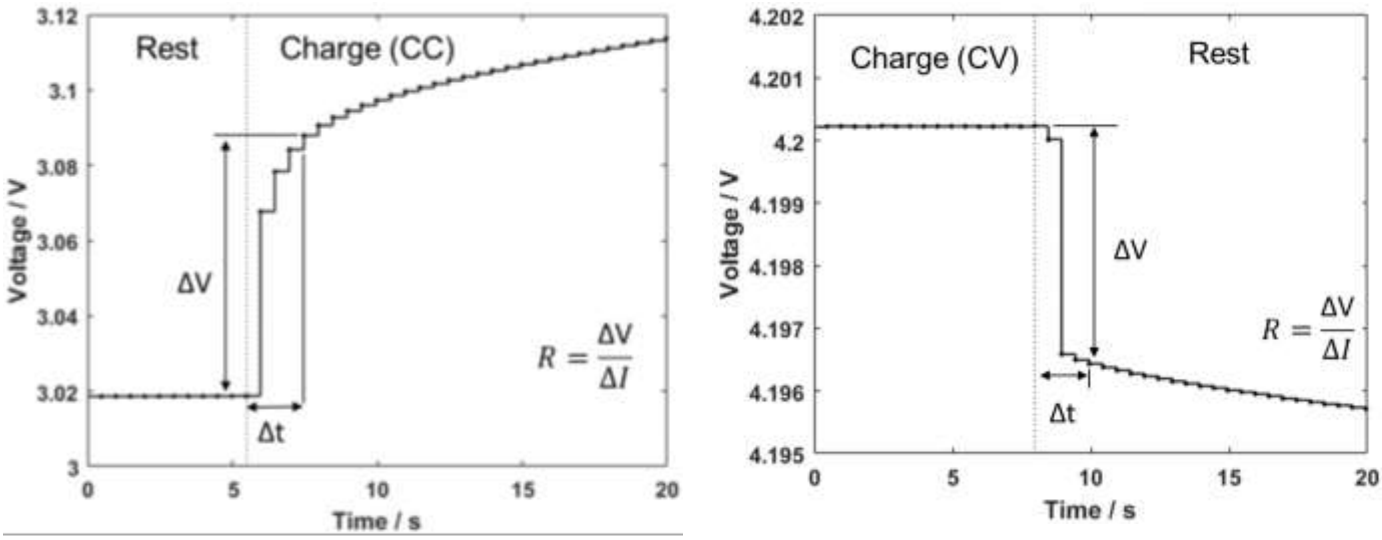
To compensate for polarisation losses shifting the voltage at which ICA peaks occur at higher C-rates, resistance-based compensation can be applied to the voltage using Ohm's law (V=IR) [7]. Current interrupt, or current pulse, is frequently used to measure resistance of ohmic devices and their connections by applying a step change in current and measuring the voltage response. The technique has been applied to batteries for onboard diagnostics were laboratory techniques such as EIS are not feasible [38], [39]. In this work, peak shift compensation is automated into the charge procedure to calculate the cell resistance using current interrupt and correct for voltage shift in the ICA. Permitting accurate ICA within charge times permissible for high power applications.

Current interrupt is performed at the start of the constant current charge and the resistance calculated as the voltage change over a fixed time period divided by the change in charge current. At the end of charge, the calculation is repeated as the constant voltage charge current cut-off threshold is reached and the cell enters a rest period. Using this method, two resistances can be calculated, one at the start of charge and one at 100% SOC and the variation in resistance with SOC accounted for using a linear interpolation. Furthermore, this method requires no additional equipment to perform and adds no time to the charging of the battery beyond the measurement time constant after charging has completed since both resistance measurements are incorporated into the charging procedure.

Unlike a conventional interrupt, where short time constants are used to isolate different dynamic effects [40], a longer time period is used to also encompass part of the kinetic polarisation effects which contribute to peak shift in ICA. The influence of time periods ranging from 0.5 s to 2.0 s on calculated resistance are discussed in section 4.2 and shown in Figure 6. Durations beyond 2.0 s were not considered, as diffusion and SOC effects will lead to overestimation of the resistance. The different steps of the peak shift correction during charge are summarised below:

1. Apply constant current charge, measure resistance using charging current interrupt.
2. Constant current charge until voltage threshold (ICA).
3. Constant voltage charge until minimum current threshold.
4. Remove charging current, measure resistance using current interrupt during voltage relaxation.
5. Compute ICA and compensate voltage by linearly interpolating resistance ($\Delta V=IR$).

Figure 1 illustrates the current interrupt technique applied to the start (a) and end (b) of a charge event. In constant current charge at the start of a charging event (a) ΔI refers to the charge current, at the end of a constant voltage charge (b) ΔI refers the charge cut-off current.



(a) (b)

Figure 1 - Current interrupt technique applied at (a) beginning of constant current (CC) charge and (b) at end of constant voltage (CV) charge.

4. Analysis

4.1. Influence of C-rate on incremental capacity analysis

4.1.1. $LiNiCoAlO_2$/graphite

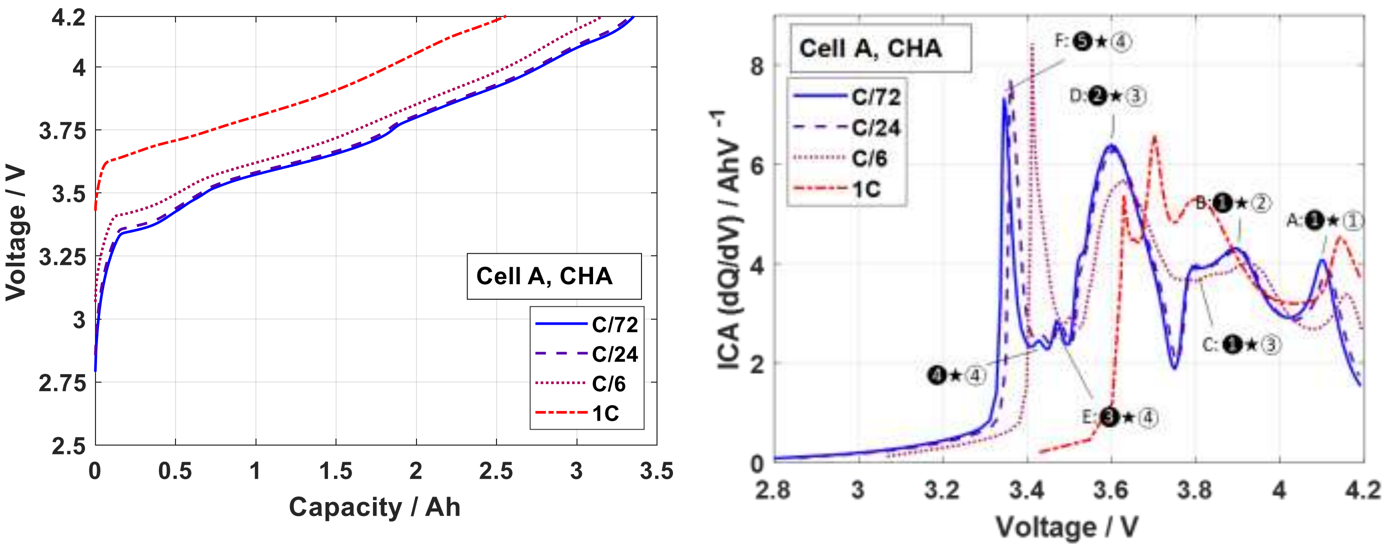
The ICA for cell A ($LiNiCoAlO_2$/graphite) under constant current charge and discharge at different C-rates is shown in Figure 2 (b) and (d) respectively. For clarity only four C-rates are shown, data from all C-rates used in this work are available in the underlying research data accompanying this paper. Peaks are labelled using the same convention as [25] and [12] with filled numbering (e.g. ❶) referring to phases in the negative electrode and unfilled numbering (e.g. ①) referring to phases in the positive electrode; ★ denotes the convolution of phases between the positive and negative electrode.

As the charge/discharge current increases, the peak locations shift to higher voltage during charge and lower voltage during discharge due to resistive and diffusion polarisation effects. In addition, as current increases, the peaks generally reduce in magnitude and become broader, losing definition and reducing the diagnostic ability of ICA. These trends are illustrated in Figure 3 which shows the evolution of peak properties with C-rate during charge. One exception to this trend is the first peak during charge (3.3 -3.4 V, Peak F), where peak magnitude increases from C/72 to C/6 before deceasing again at higher currents (Figure 3c). ICA on an NCA half-cell performed by [41] shows a similar sharp peak at the onset of charge, suggesting this behaviour is likely due to de-intercalation at the positive electrode. Half-cell tests on the same positive electrode chemistry by [12] also showed an increase in the first charge peak magnitude from C/40 to C/5, followed by a reduction at 1C. One potential mechanism is that increased polarisation effects at low SOC reduce the amount of charge transferred in the low voltage region (2.8-3.3 V) prior to the first peak at higher C-rates, this then increases the amount of charge transferred during the first voltage plateau, increasing the peak magnitude.

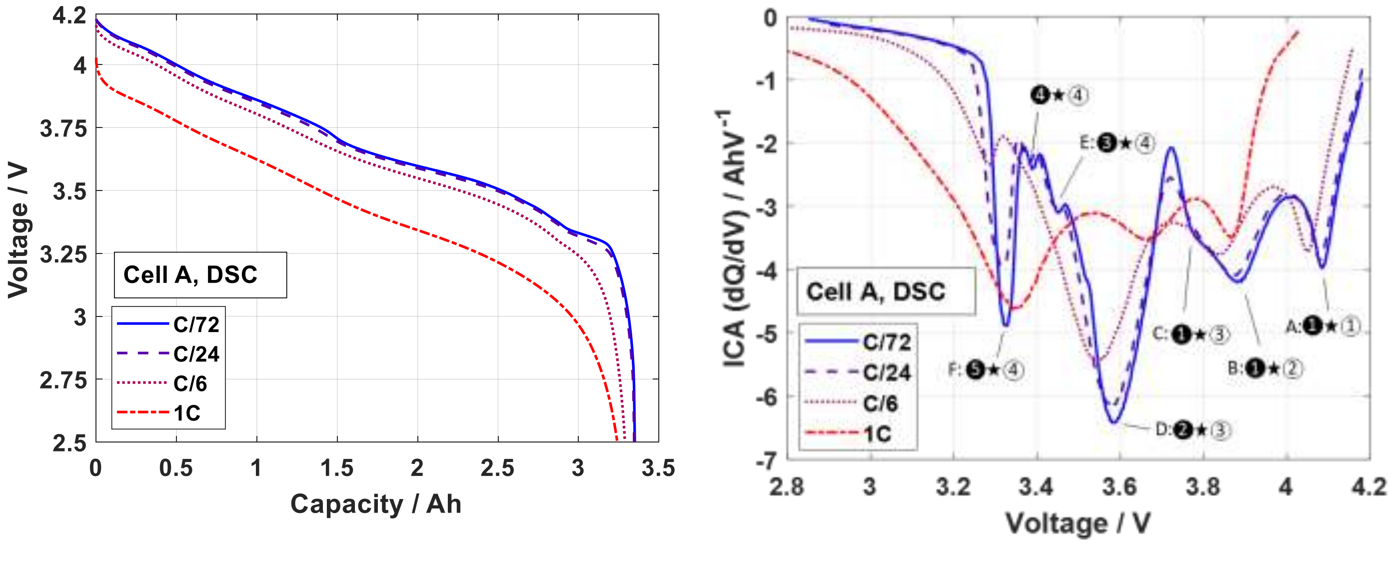
Table 2 demonstrates how the number of identifiable peaks change with C-rate using the fitting method discussed in section 3. As C-rate increases, the number of peaks in the ICA reduces from a maximum of six peaks at C/48 charge to three peaks at 1C charge and discharge. High charge/discharge rates increase the concentration gradients in the electrodes and reduce the relaxation time available for phases to equalise, increasing the probability of inhomogeneous lithium distribution [32]. Inhomogeneous lithium distribution can lead to multiple different intercalation phases co-existing across the electrode thickness, broadening and reducing the heights of the peaks in the ICA. As current continues to increase, adjacent peaks will combine, reducing the number of identifiable peaks, as seen in Figure 2.

Changes in temperature caused by Joule heating at high C-rates will also influence the cell behaviour, effecting the accuracy of the ICA [19]. Whilst the test chamber temperature was maintained to 25.0±1.0 °C throughout all tests, the cell was not actively cooled. During the 1C charge of cell A, cell surface temperature increased from the chamber temperature to a peak of 38.8 °C at the end of constant current charge. During the C/72 charge, temperature remained within the ±1.0 °C tolerance of the chamber.

Only five peaks were identified during the C/72 cycle, one less than at C/48; this is likely because the C/72 cycle was conducted as the last test and the small amount of degradation from use has influenced the shape of the ICA. Overall, very little difference was seen in the ICA performed at C/72, C/48 and C/24, suggesting that the widely used rates of C/24 and C/25 represent a good compromise between accuracy and time for ICA in a laboratory setting. In an application-based setting where faster charging is required, such as an electric vehicle, C/6 represents a reasonable compromise between ICA accuracy and time available for uninterrupted constant current charge; for example, overnight or workplace charging of an electric vehicle. The validity of this will be evaluated using ageing tests in section 4.3.

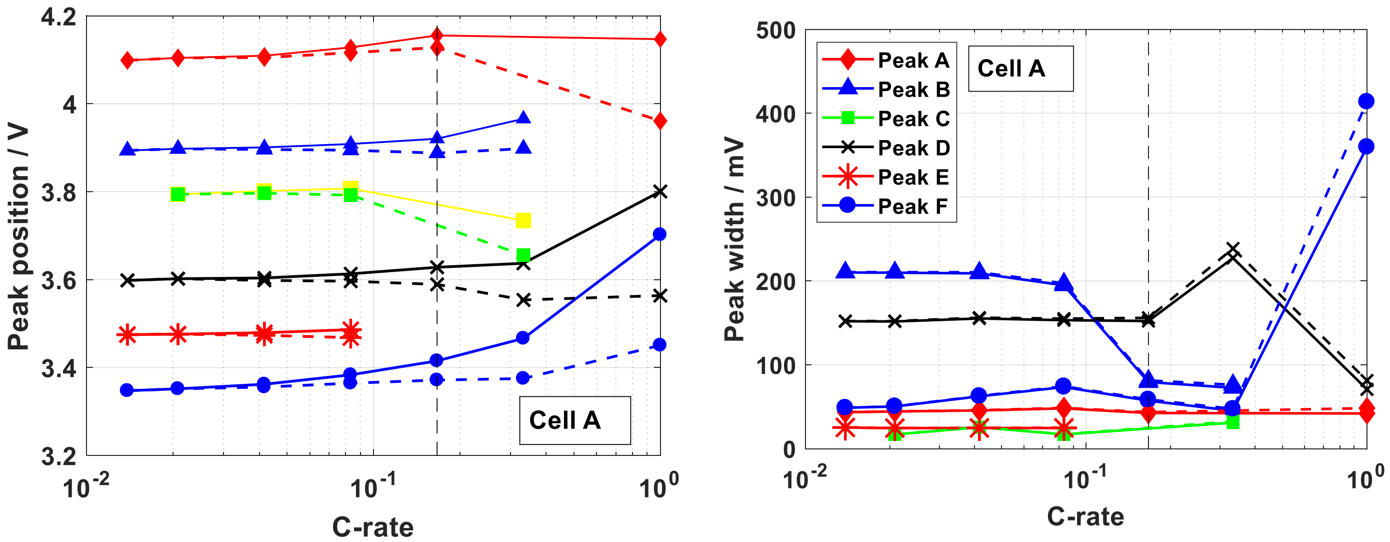


(a) (b)



(c) (d)

Figure 2 – $LiNiCoAlO_2$/graphite cell cycled at different rates (a) charge (CHG) profile, (b) charge ICA, (c) discharge (DSC) profile and (d) discharge ICA (inverted y-axis). Peak labelling in (b) and (d) corresponds to C/72.



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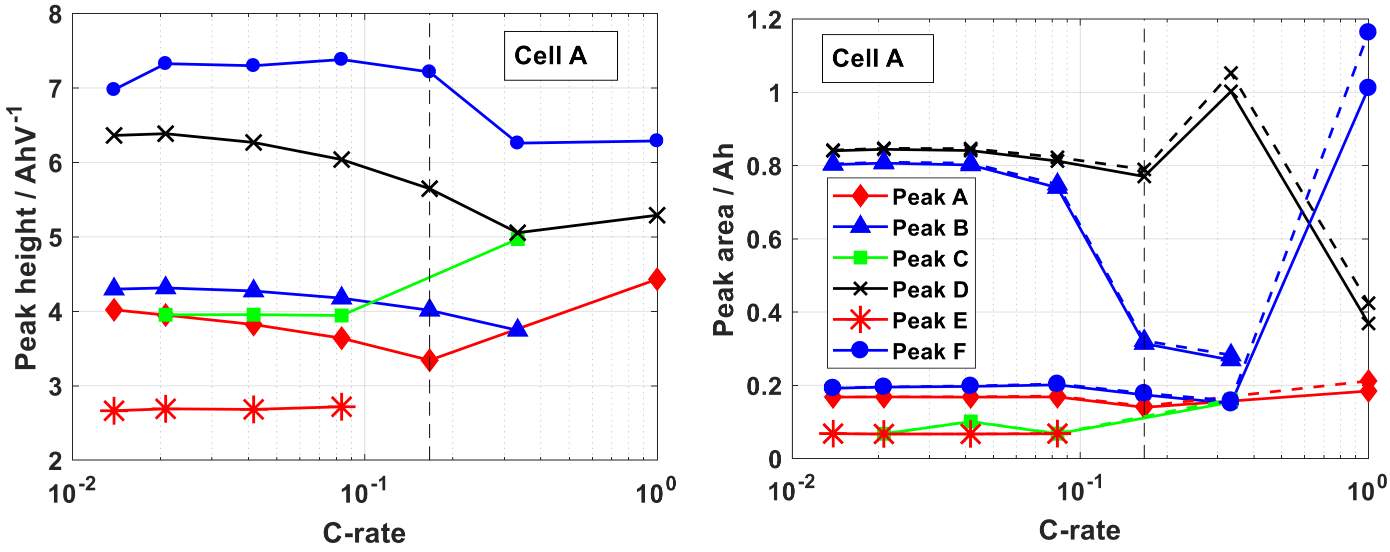


(c) (d)

Figure 3 - $LiNiCoAlO_2$/graphite (cell A) peak properties as a function of C-rate during charge (a) Peak position, (b) Peak width, (c) Peak height and (d) Peak area. Dashed lines represent IR corrected values. Vertical dashed line to illustrate C/6.

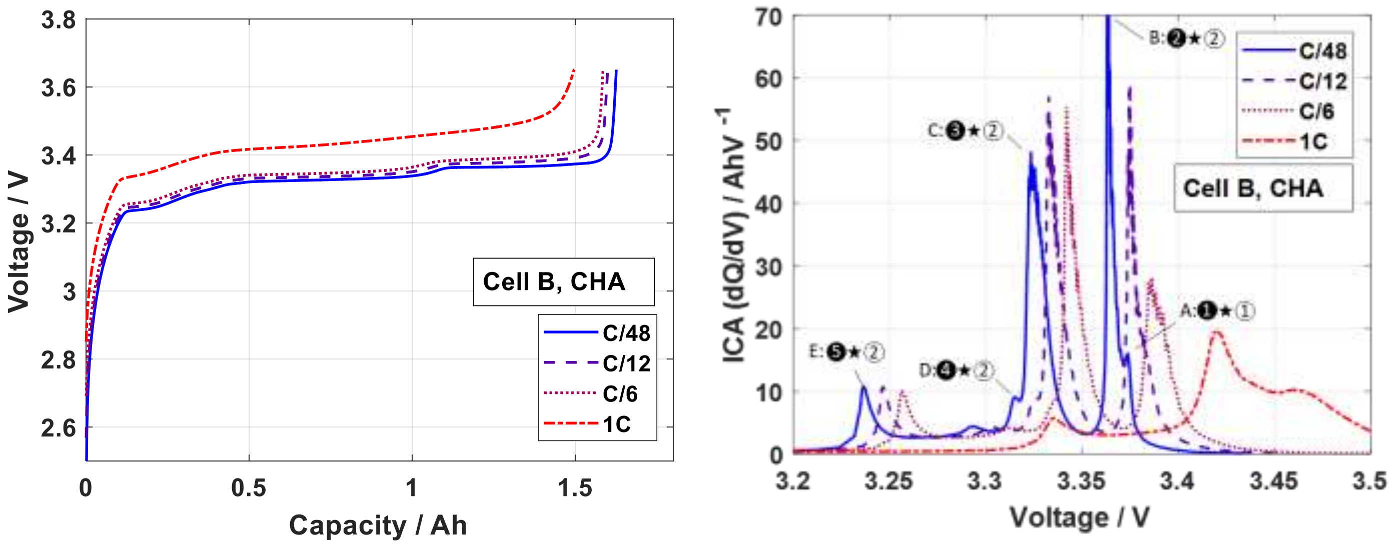
C-rate		C/72	C/48	C/24	C/12	C/6	C/3	C/2	C/1
Cell A	Charge	5	6	6	6	4	4	-	3
	Discharge	4	4	4	4	4	3	-	3
Cell B	Charge	-	5	4	4	4	3	3	3
	Discharge	-	5	5	5	4	3	3	2

Table 2 - Number of unique peaks identified in ICA as a function of C-rate

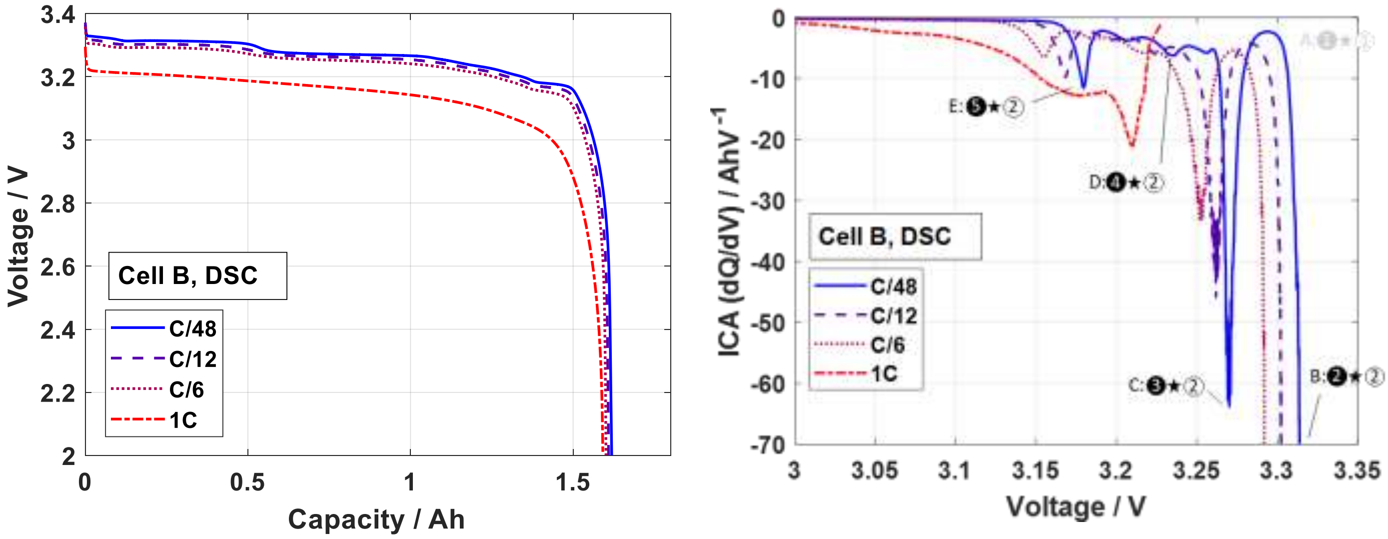
4.1.2. $LiFePO_4$/graphite

$LiFePO_4$ is widely used as a positive electrode in ICA studies since 95% of the capacity change occurs across a single voltage plateau. Through identifying the ICA peak caused by the transition from de-lithiated $FePO_4$ to fully lithiated $LiFePO_4$ it is reasonable to conclude that the other peaks are due phase transformations in the negative electrode [9].

Figure 4(b) and (d) show the charge and discharge ICA respectively for cell B at different C-rates. As with cell A, the same general trends of peak shifting and broadening at higher currents is observed but condensed into a narrower voltage range with more distinct peaks of higher magnitude. The graphite negative electrode undergoes five distinct stages from C_6 to LiC_6 as Li^+ is intercalated between carbon layers [42]. At C/48, five peaks can be identified in the ICA of the $LiFePO_4$/graphite cell, likely corresponding to the four phase transformations in negative electrode and one phase transformation in the positive electrode. Table 2 shows that as C-rate increases, the number of identifiable peaks for cell B reduce. As with cell A, C/24 shows little loss of information compared to C/48 and C/6 represents a reasonable compromise between time and accuracy for applications such as electric vehicle charging, with no loss of peaks during charge compared to C/24.



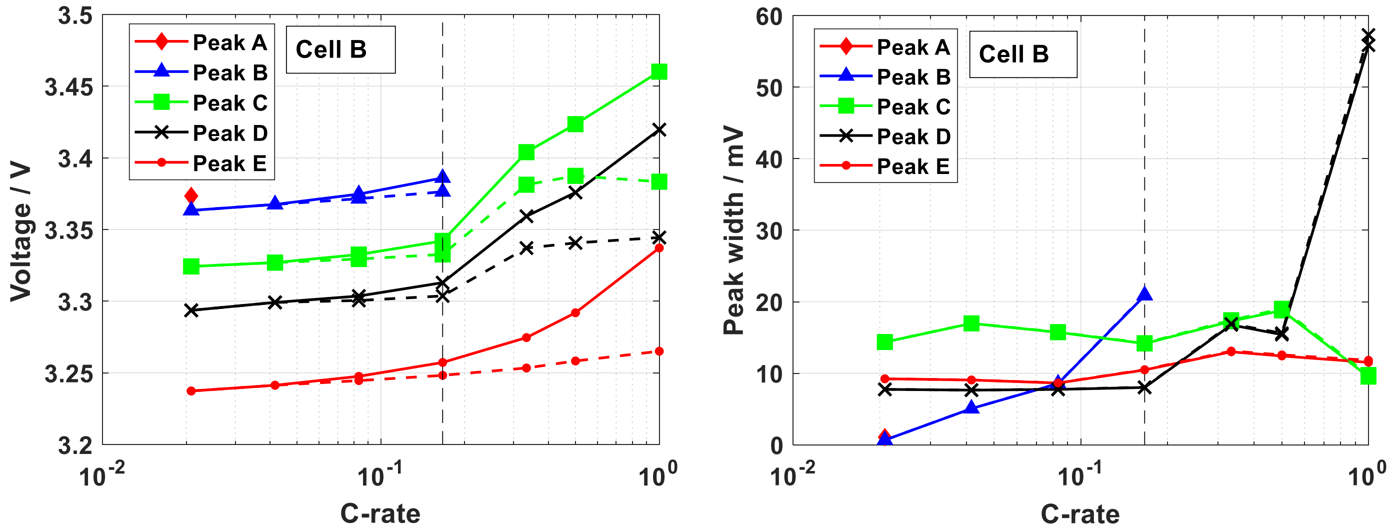
(a) (b)



(c) (d)

Figure 4 - LiFePO₄/graphite cell cycled at different rates (a) charge (CHA) profile, (b) Charge ICA, (c) discharge (DSC) profile and (d) discharge ICA (inverted y-axis). Peak labelling in (b) and (d) corresponds to C/48.

Comparing the peak properties in Figure 5, peak B shows the greatest change in shape at higher C-rates, significantly reducing in height, widening and increasing in area. This peak jointly corresponds to the deintercalation of $LiFePO_4$ to $FePO_4$ and one of the negative electrode intercalation stages. Higher C-rates imply greater concentration gradients within the electrodes, leading to less pronounced peaks with greater width. As with Cell A, the remaining peak properties show a consistent trend from C/48 up to C/6 (vertical dashed line in Figure 5). At charge currents above C/6, peak properties do not reflect the behaviour seen at lower currents, especially in the heights and area of peaks C and D in Figure 5. The maximum cell temperature during the 1C discharge was 30.7 °C.



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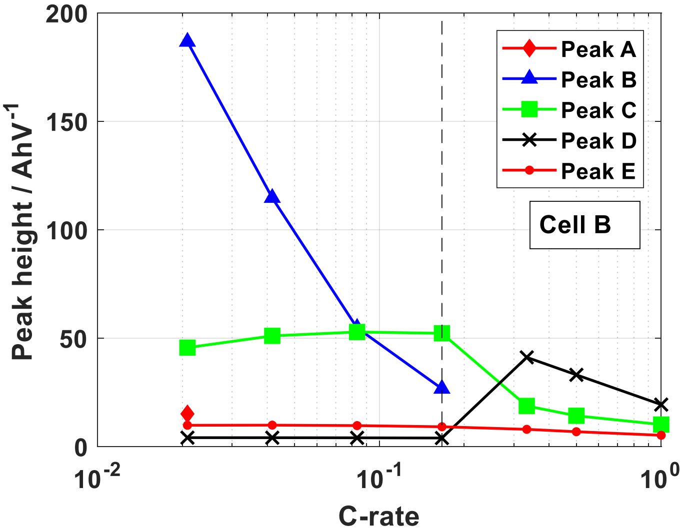
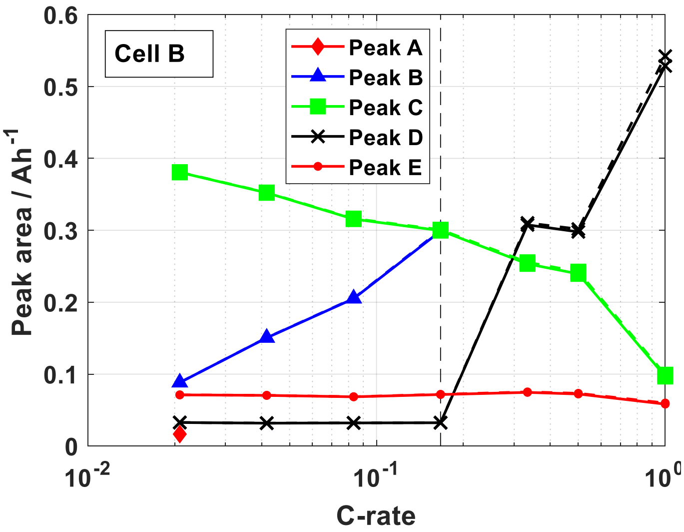


(c)



(d)

Figure 5 - $LiFePO_4$/graphite cell peak properties as a function of C-rate during charge (a) Peak position, (b) Peak width, (c) Peak height and (d) Peak area. Dashed line is IR corrected. Vertical dashed line to illustrate C/6.

Neither cell A nor B are optimised for high power output over high energy. Reduced resistances and thinner electrodes seen in power optimised cells may mean that ICA can be performed at higher C-rates compared to energy optimised cells, provided concentration gradients within the electrode do not become too large and temperature increase is controlled.

4.2. Polarisation compensated ICA

Two different methods of determining cell resistance for peak shift compensation were explored, current interrupt, and EIS. Figure 6 compares the resistances obtained to the measured peak shift. The measured peak shift is determined as the voltage shift of peaks at different C-rates divided by the current difference (R = V/I), performed on cell A at C/6 charge. Due to the varying shape and change in number of the peaks, it is not possible to determine an effective resistance for every peak.

It can be seen from Figure 6 that the effective resistance increases with reducing SOC, with the peak shift calculated resistance showing the most significant increase. As such, any peak shift correction should also account for the change in resistance with voltage. The EIS high frequency intercept underestimates the amount of voltage shift required at all SOC levels. This is because the high frequency intercept, which typically occurs around 1 kHz, only accounts for the electrical resistance and electrolyte ionic conductivity, not diffusive polarisation losses which take longer to establish. EIS at 1.0 Hz shows an increase in measured real resistance with reducing frequency, especially at low SOC, but not to the extent of the peak shift. Further reducing the frequency to 0.1 Hz is seen to give a result more in line with that observed in the peak shift, showing a large resistance increase between 3.3 and 3.6 V. However, despite some lab based studies, the application of low cost on-board EIS diagnostics is still not well developed [23] [43], and the time taken to obtain accurate measurement at the start and end of charging would impact the vehicle charging time.

To account for polarisation losses and better represent the true behaviour of the cell, current interrupt resistance, measured from the voltage change at different time constants after the charge current has been applied/removed (section 3.2), is shown in Figure 6. As the time constant increases, the amount of diffusive polarisation accounted for increases, however excessive time after applying a current could also account for the voltage increase during charge, overestimating the resistance. The influence of time constant at low SOC is significant as diffusive polarisation dominates in this region [40]. At high SOC the time constant makes little difference to the resistance.

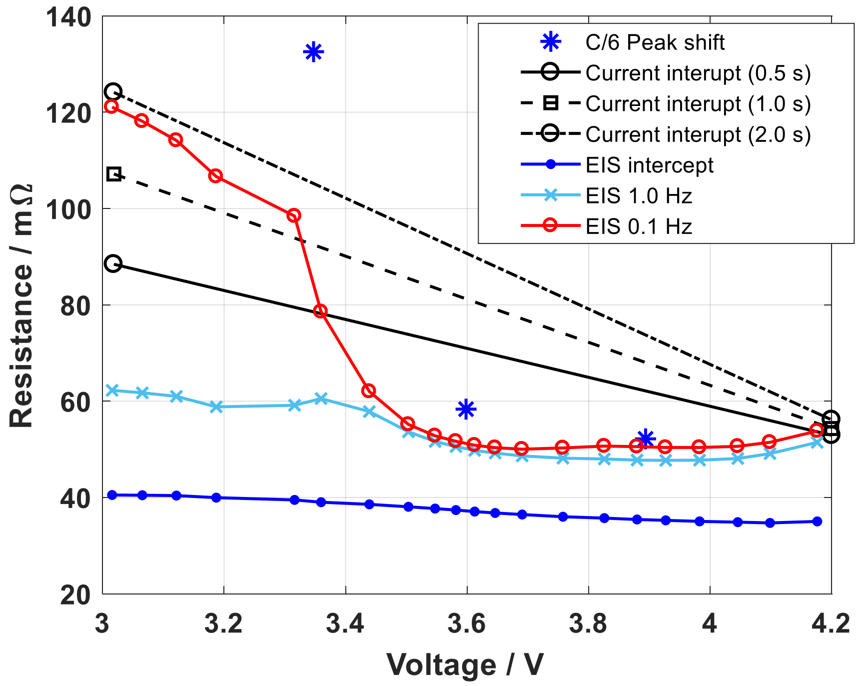


Figure 6 - Measured resistance as a function of voltage for cell A

Based on these results, current interrupt with a time constant of 1.0 s is chosen for the remainder of this work as it takes into consideration a sufficient amount of the polarisation losses whilst not taking an excessive time that may lead to state of charge effects being considered. A sample period of 1.0 s is also likely to be wholly divisible by the battery management sample frequency (i.e. 1 Hz, 2 Hz etc), reducing measurement error between consecutive cycles. Resistances are measured at the onset of charge and at the end of charge (after the minimum current threshold) as part of the charging procedure described in section 3.2 and illustrated in Figure 1. Linear interpolation is then used to express the resistance as a function of voltage throughout the SOC range.

Considering the robustness to error in the resistance calculation, a ±0.25 s error in the current interrupt time constant would lead to a 7.48% and 2.39% maximum error in the calculated resistance value for the lowest and highest voltage peak location of cell A respectively. At C/6, this equates to a maximum voltage shift error of 3.85 mV at the low voltage peak (peak F) and 0.78 mV at the high voltage peak (peak A).

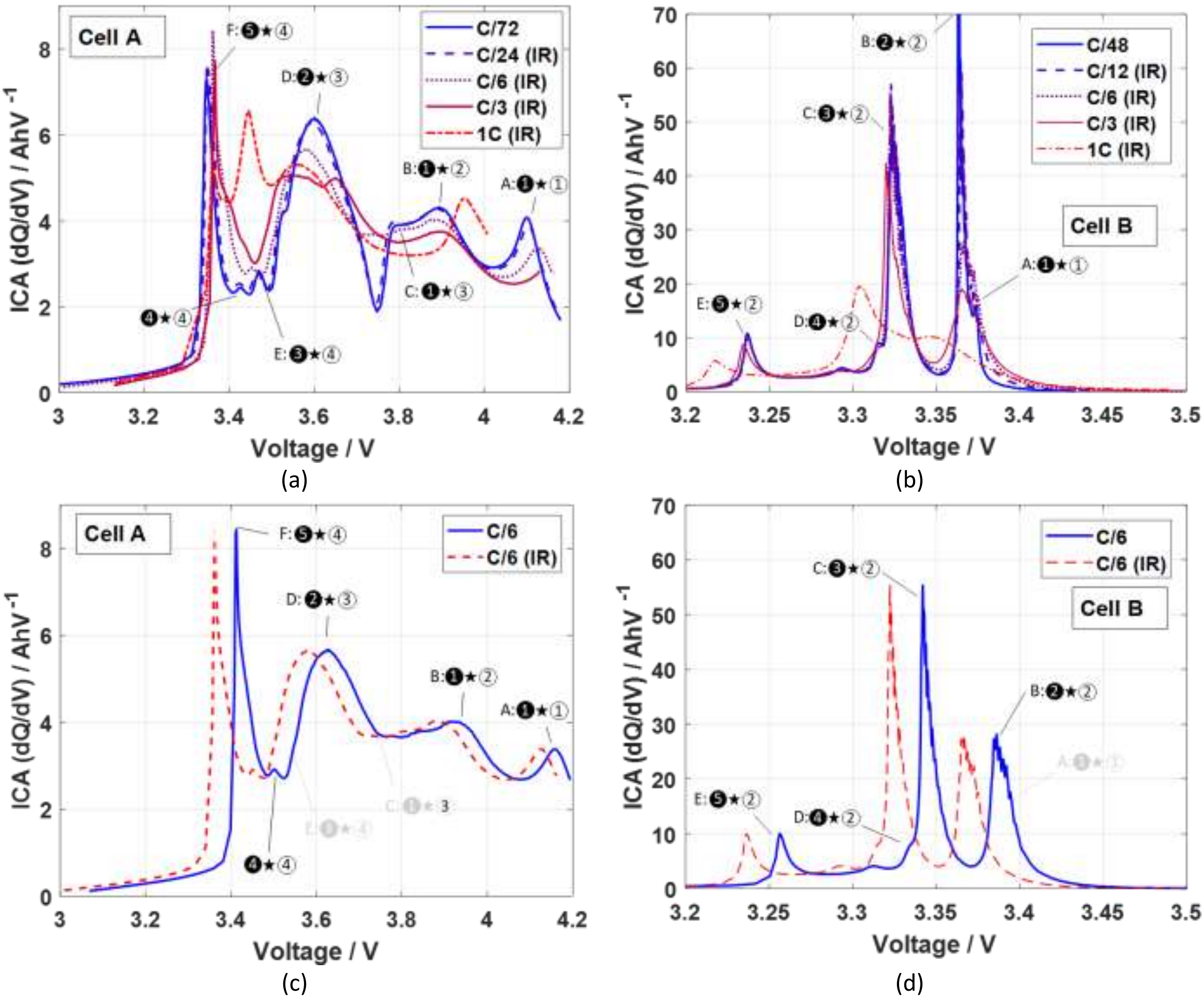


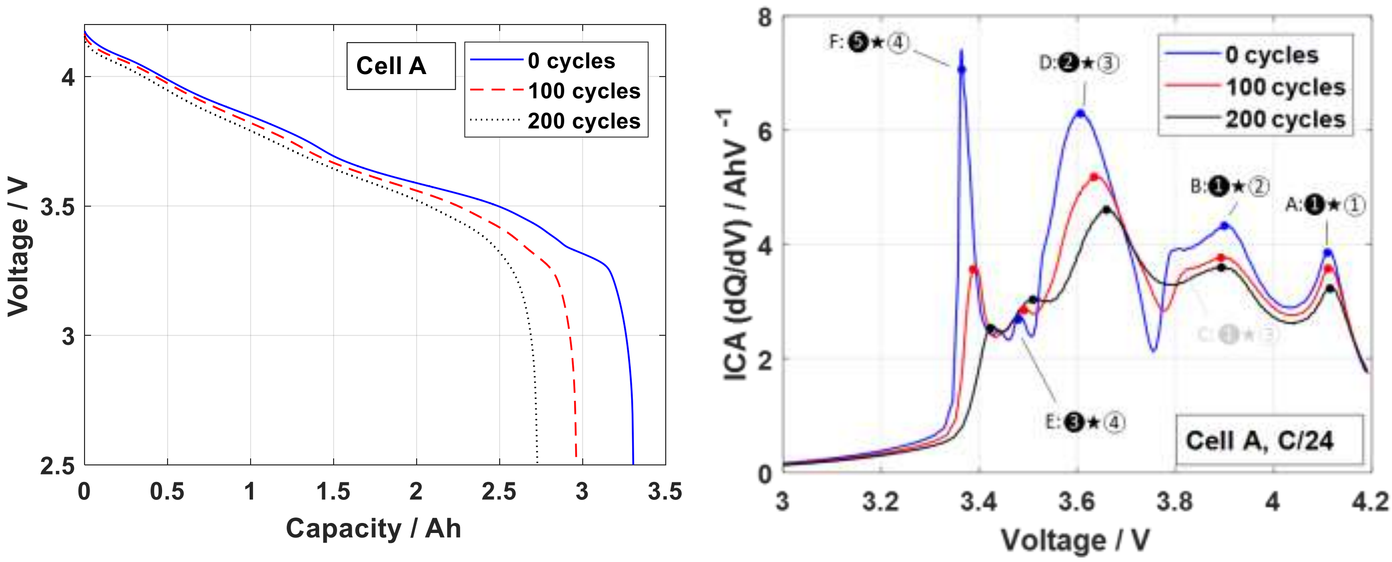
Figure 7- ICA with polarisation compensation applied (indicated with IR in legend) (a) Cell A and (b) Cell b

Figure 7 shows the ICA of cell A and B with polarisation compensation applied, demonstrating the effectiveness of the proposed method. Correcting for peak shift also better illustrates how the shape of ICA changes with C-rate. For cell A, between C/72 and C/3 the general shape of the ICA is preserved but with reductions in peak magnitude and a loss of definition. At 1C a new peak is seen to form at 3.45 V, whereas the same voltage at lower C-rates coincides with a trough. The distinct change in shape from the quasi-OCV tests at C/24 implies that conventional correlations between voltage and phase transformations cannot be applied to such high C-rates for this cell, even with polarisation compensation. A similar trend is also seen in cell B, where the location of the three main peaks is maintained up to C/3, but a large shift is seen to occur at 1C.

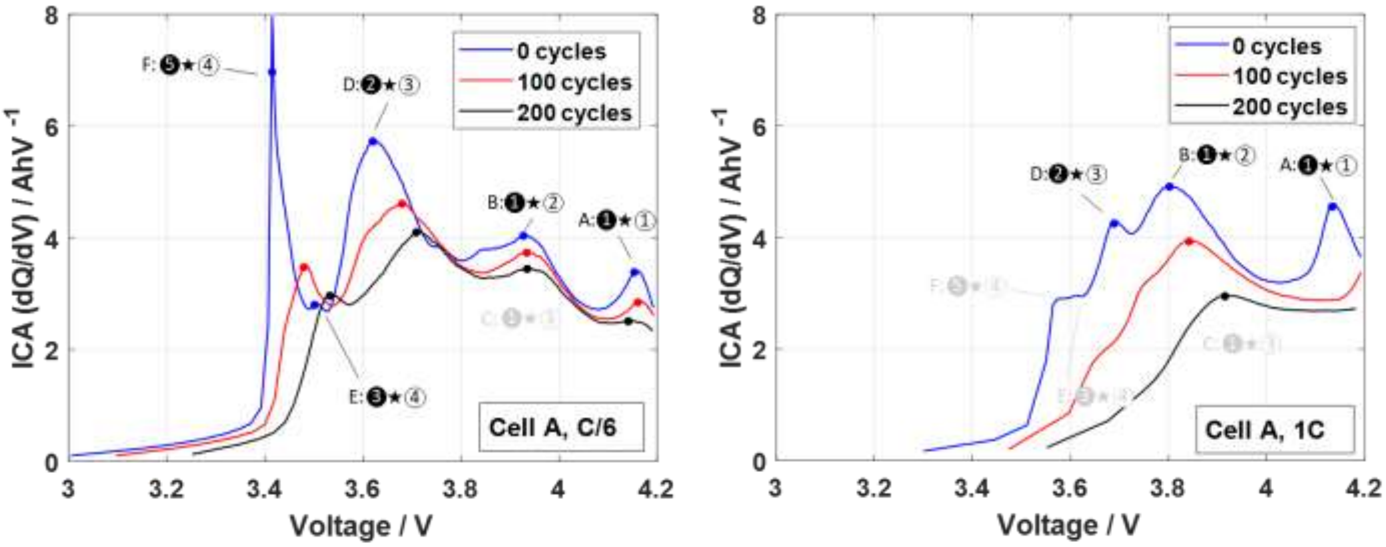
The influence of the proposed current interrupt method on peak location, width and area is shown in Figure 3 and Figure 5 for cell A and B respectively. For a C/6 charge using polarisation compensation, the peak voltages are within 0.59% and 0.10% of C/48 for cell A and B respectively compared to 1.90% and 0.67% without compensation. This technique therefore represents an accurate method to reduce the time taken for ICA to be performed with no additional equipment requirements and minimal computation. Polarisation compensation is also seen to have a small influence on the peak area and peak width. This occurs because the peak shift method accounts for resistance change across a single peak using a linear interpolation between the resistance at the beginning and end of charge as shown in Figure 6. The change in peak width then affects the calculated peak area, as shown in Figure 3(d) and Figure 5(d). Polarisation compensation allows for peak voltage location at higher C-rates to be better correlated to peak locations obtained from laboratory-based studies at lower C-rates or from steady state simulations. This can aid with the identification of phase change behaviour and degradation mechanisms, especially when peaks are close in voltage and shape. Furthermore, polarisation compensation helps to remove the contribution of ORI to the ICA, improving the ability to diagnose LAM or LLI, especially when these degradation modes result in a voltage shift of specific peaks.

4.3. Incremental capacity analysis for degradation analysis

One of the primary uses of ICA is to analyse the SOH over the useable lifetime of the battery. To analyse the ability of ICA at different C-rates to characterise SOH with and without a polarisation compensation as a diagnostic tool, cell A was aged over repeated 1C cycles with characterisation cycles at C/24, C/6 and 1C every 100 cycles. Figure 8 shows the C/24 discharge curve and ICA throughout the lifetime of the cell and different C-rates without polarisation compensation. Peak locations obtained from peak fitting are highlighted using dots in Figure 8 (b, c and d).



(a) (b)



(c) (d)

Figure 8- Influence of C-rate on ICA for a cycled cell without polarisation compensation applied (a) discharge curves at C/24, (b) ICA at C/24, (c) ICA at C/6, (d) ICA at 1C

The cell is seen to lose 10.4% and 17.5% of its capacity after 100 and 200 cycles respectively, which is very close to the 18.0% predicted by the manufacturer's datasheet after 200 cycles under similar conditions [44]. The cell resistance at 100% SOC, measured using current interrupt, increased by 22.8 mΩ and 37.3 mΩ after 100 and 200 cycles respectively relative to the fresh cell. Comparing the three different ICA rates, C/24 and C/6 demonstrate a very similar trend in the evolution of peak shape and location over time, whereas the 1C ICA loses a significant amount of definition and is reduced to only one identifiable peak after only 100 cycles. The height, voltage (with and without polarisation compensation), width and area of each peak throughout the degradation study is shown in Table 3, where C/24 is taken as the reference current for polarisation compensation. The change in peak properties with cycling are shown in Figure 9 where the solid line represents C/24 and the dashed line C/6. Properties at 1C are not shown for simplicity since only one peak was identified, likewise peak D is omitted since it was not identified at C/6 after aging.

C-Rate	Peak	0 cycles					100 cycles					200 cycles				
		H	V	IR	W	A	H	V	IR	W	A	H	V	IR	W	A
C/24	A	3.85	4.11		43	159	3.57	4.11		49	188	3.23	4.11		50	123
	B	4.32	3.90		199	769	3.77	3.89		153	555	3.59	3.89		87	352
	D	6.29	3.61		149	825	5.18	3.63		186	739	4.61	3.66		407	1208
	E	2.68	3.48		20	53	2.85	3.49		17	55	3.03	3.51		22	82
	F	7.06	3.36		49	178	3.56	3.39		28	49	2.53	3.42		17	16
C/6	A	3.39	4.15	4.12	51	167	2.85	4.16	4.13	37	95	2.51	4.14	4.10	29	47
	B	4.04	3.93	3.89	92	363	3.74	3.93	3.90	78	282	3.45	3.93	3.89	65	197
	D	5.73	3.62	3.58	152	757	4.61	3.68	3.64	205	767	4.10	3.71	3.67	376	1012
	E	2.80	3.50	3.46	10	28	-	-	-	-	-	-	-	-	-	-
	F	6.97	3.41	3.37	56	163	3.48	3.48	3.44	39	81	2.97	3.53	3.49	34	50
1C	A	4.55	4.13	3.95	57	250	-	-	-	-	-	-	-	-	-	-
	B	4.91	3.80	3.62	223	994	3.93	3.84	3.81	151	493	2.95	3.91	3.88	102	221
	D	4.25	3.69	3.50	23	99	-	-	-	-	-	-	-	-	-	-

Table 3 - Peak information from ICA of cycled Cell A at different rates: H = Peak height (AhV^{-1}), V = Peak voltage (V), IR = polarisation compensated peak voltage (V), W = Peak width (mV) and A = Peak area (mAh)

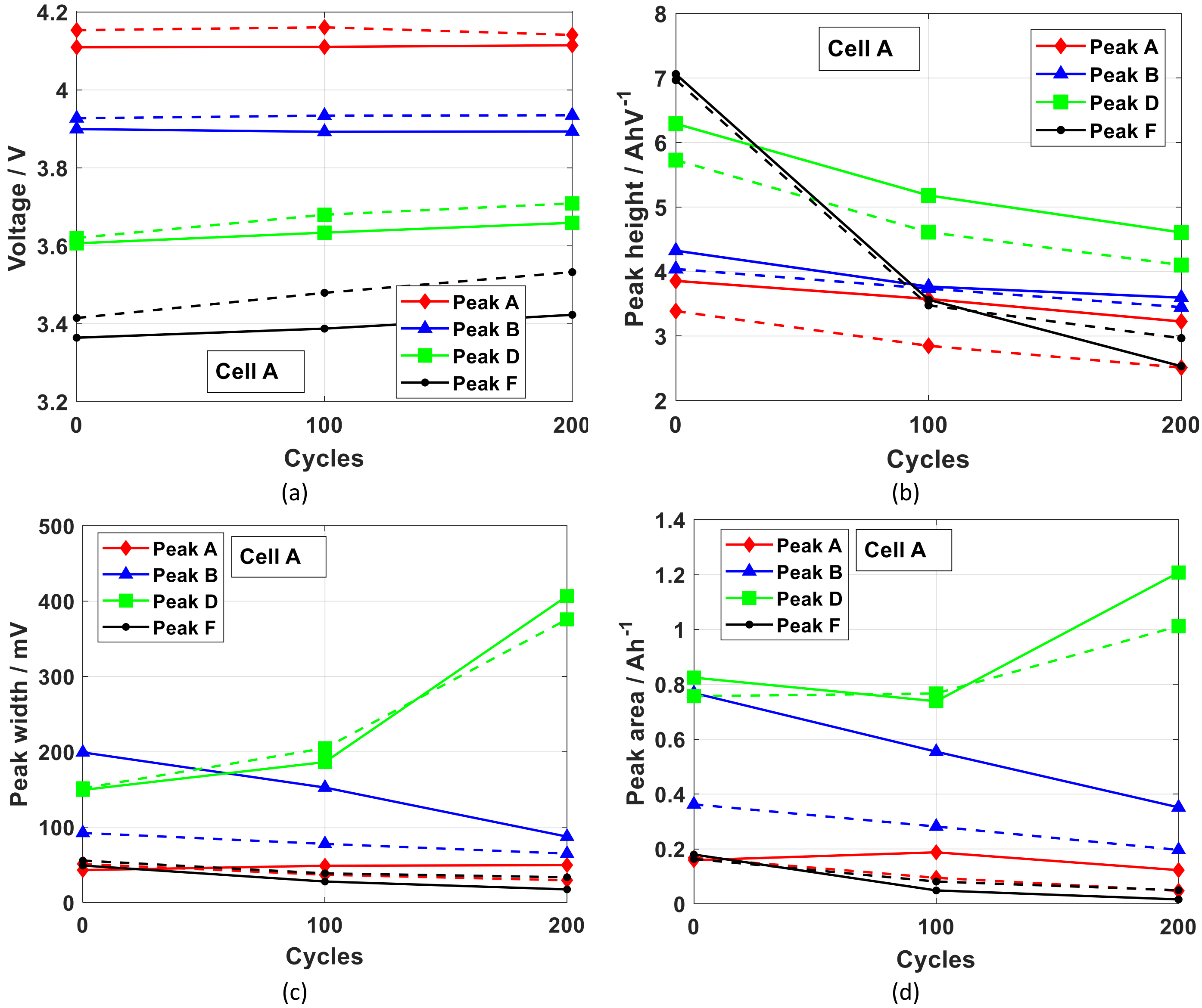


Figure 9 - Change in peak properties after 200 1C cycles, solid lines from C/24 ICA and dashed lines from C/6 ICA. (a) Peak position no IR correction applied, (b) Peak width, (c) Peak height and (d) Peak area.

To relate the change in ICA peak to the underlying degradation modes, and investigate the influence of C-rate, the relationships observed by Devie and Dubarry [12] for the same $LiNiCoAlO_2$/graphite 18650 cell will be used. This technique used half-cell data from the cell to deconstruct the whole cell ICA into contributions from the separate electrodes. They then analysed the half-cell data in the Alawa toolbox to observe how degradation mechanisms will affect the whole cell ICA response [4]. This technique considers the influence of LAM, LLI and ORI. LAM is further split into the positive electrode (LAM_{Pe}) and negative electrode (LAM_{Ne}), during lithiation (LAM_{li})and de-lithiation (LAM_{de}). For example, LAM_{liPe} corresponds to loss of active material during lithiation of the positive electrode.

LAM_{dePe} and LAM_{liPe} can be separated from other degradation modes by a reduction in height of peak A and B [12], both of which can be seen to occur at C/24 and C/6. LAM_{dePe} is more prominent than LAM_{liPe} in the aging tests as the voltage of peaks A and B are preserved, combined with a large reduction in height of peak F. This is observed at both C/24 and C/6 where the height of peak F (3.36-3.53 V) reduces significantly compared to other peaks (Figure 8b). However, this diagnosis cannot be made at 1C as peak A is only identified for the fresh cell, disappearing after 100 cycles, and peak F was not identified as a unique peak.

LLI is identified as a shift in peak F to higher voltages combined with a reduction in height of peak D [12] which can be clearly seen in the C/24 (Figure 8b) and C/6 (Figure 8c). Once again, this behaviour cannot be seen in the 1C ICA (Figure 8d) as peak D was only identified in the fresh cell.

LAM_{liNe} has a similar ICA signature to LLI with the difference being that peak C remains unchanged for LLI but becomes separated from peak B in the case of LAM_{line} [12]. Since peak C could not be identified after aging at any of the C-rates tested, it is not possible to separate LLI and LAM_{liNe} from the full cell ICA. LAM_{deNe}, exhibited as a disappearance of the high voltage peak A, cannot be seen in Figure 8b, indicating that this degradation mode does not have a significant influence on the observed capacity loss. ORI can be directly determined from the current interrupt method discussed in section 4.2.

Comparing ICA at C/24 (Figure 8b) and C/6 (Figure 8c), there are several identifiable differences with aging. Peak E (3.48-3.53 V), the narrowest of all the peaks could not be identified in the C/6 ICA from 100 cycles onwards and peak A (4.10-4.15 V) was seen to lose prominence with aging at C/6 whist at C/24 only a small reduction in height was observed. The latter observation, occurring at the end of charge, may be due to concentration gradients within the electrolyte or electrodes that occur at the higher C/6 rate.

Overall, the main degradation modes identified in the aged cell are LAM_{dePe} and either LAM_{liNe} or LLI. Whilst the peaks are more defined at C/24, the trends required to identify the degradation modes discussed can still be seen at C/6. At 1C, the loss of peak definition and number of identifiable peaks makes ICA unsuitable as an effective diagnostic tool for the cell chemistry tested. It is anticipated that the findings from this cell will extend to aging studies of other cells with different chemistries and form factors at similar C-rates. Cells with simpler phase change behaviour, such as $LiFePO_4$ and power optimised cells may be capable of ICA at higher C-rates after aging. Consideration should also be given to additional rest periods prior to constant current charging in large format pouch cells due to the increased ionic pathway distance between the anode overhang and cathode [33]. Depending on the battery condition prior to charging, it may take multiple days of resting for homogeneous lithium distribution to occur and to separate reversible from irreversible capacity loss. The relationship between reversible and irreversible capacity loss will also change throughout the lifetime of the cell [45]. Fully accounting for inhomogeneity and reversible effects through extended rest periods may lead to a shift in the negative electrode voltage profile relative to the positive electrode which in turn could influence the ICA shape. In the current work, the same rest period was used prior to each cycle and the difference between reversible and irreversible capacity reduction not considered.

4.4. Polarisation compensated ICA on aged cell

Under the aging test, the addition of polarisation compensation is seen to reduce the voltage difference between the C/6 and C/24 ICA at all identifiable peaks compared to without compensation (Table 3), demonstrating the viability of this method throughout the operating lifetime of the cell.

Figure 10(a-c) shows how the different C-rates perform with polarisation compensation at C/24, C/6 and 1C respectively at different stages of ageing. The results show that C/6 provides a good compromise between maintaining the accuracy of ICA at slow C-rates whilst satisfying the time constraints of conventional charging procedures for applications such as electric vehicles. It can also be seen that ICA must be performed at the same C-rate throughout the lifetime of the cell to obtain meaningful information about peak height and width evolution. For example, a corrected C/6 after 100 cycles cannot be compared to a C/24 after 200 cycles to measure SOH. The application of polarisation compensation also indicates that the sole identifiable peak in the 1C ICA of the aged cell corresponds to peak D in the C/24 and C/6 ICA.

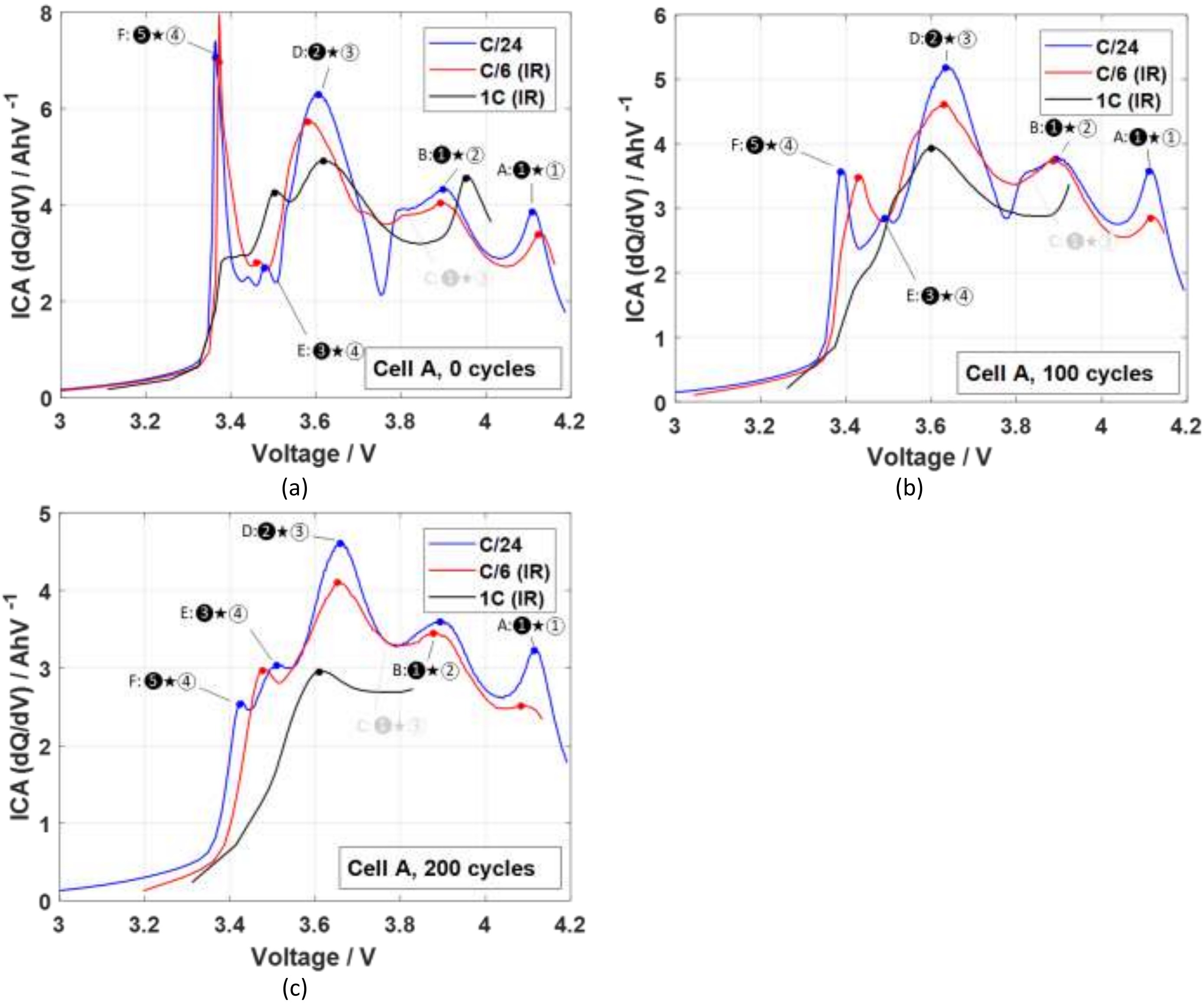


Figure 10 – ICA at different C-rates with polarisation compensation after (a) 0 cycles, (b) 100 cycles and (c) 200 cycles

Conclusions

The influence of current on the accuracy of ICA as a diagnostic tool has been studied quantitively on two different lithium-ion battery chemistries through identification of peak location, height, width and area. Results show that as current increases and the cell deviates further from the open circuit voltage the number of identifiable peaks reduce, peaks become lower in magnitude, broader and shift in voltage. These effects were seen to be more prominent in a cell that had undergone ageing, demonstrating the need to consider ageing when selecting an appropriate C-rate for ICA. This work represents the first quantitative study linking rate-dependency to peak properties in both unaged and aged lithium-ion cells.

Based on these findings, a new method has been presented to correct peak shift in ICA at higher currents by using current interrupt to estimate cell resistance with linear interpolation. Using this technique, ICA obtained over a 6-hour charge (C/6) is shown to predict peak location voltage to within 0.59% of a 48-hour (C/48) quasi steady state charge. This method also helps to remove resistance induced peak shift in aged cells from ICA at higher currents, simplifying the diagnostics of other degradation modes.

Comparison of ICA performed at different C-rates shows that lower C-rates provide more defined peaks and are therefore better for SOH diagnostics. However, reliable identification of degradation mechanisms can still be performed up to C/6 using existing diagnostic tools. This work demonstrates that ICA can be obtained over a conventional overnight or workplace electric vehicle charging period for use in SOH diagnostics for lithium-ion batteries.

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References

[1] H. Berg, *Batteries for electric vehicles: Materials and electrochemistry*. 2015.

[2] M. Berecibar, I. Gandiaga, I. Villarreal, N. Omar, J. Van Mierlo, and P. Van Den Bossche, “Critical review of state of health estimation methods of Li-ion batteries for real applications,” *Renew. Sustain. Energy Rev.*, vol. 56, pp. 572–587, 2016.

[3] X. Han, M. Ouyang, L. Lu, J. Li, Y. Zheng, and Z. Li, “A comparative study of commercial lithium ion battery cycle life in electrical vehicle: Aging mechanism identification,” *J. Power Sources*, vol. 251, pp. 38–54, 2014.

[4] M. Dubarry, C. Truchot, and B. Y. Liaw, “Synthesize battery degradation modes via a diagnostic and prognostic model,” *J. Power Sources*, vol. 219, pp. 204–216, 2012.

[5] L. Balewski and J. P. Brenet, “A New Method for the Study of the Electrochemical Reactivity of Manganese Dioxide,” *Electrochem. Technol.*, vol. 5, pp. 527–531, 1967.

[6] M. Dubarry, V. Svoboda, R. Hwu, and B. Yann Liaw, “Incremental Capacity Analysis and Close-to-Equilibrium OCV Measurements to Quantify Capacity Fade in Commercial Rechargeable Lithium Batteries,” *Electrochem. Solid-State Lett.*, vol. 9, no. 10, p. A454, 2006.

[7] M. Dubarry and B. Y. Liaw, “Identify capacity fading mechanism in a commercial LiFePO4 cell,” *J. Power Sources*, vol. 194, no. 1, pp. 541–549, 2009.

[8] M. Dubarry *et al.*, “Evaluation of commercial lithium-ion cells based on composite positive electrode for plug-in hybrid electric vehicle applications. Part II. Degradation mechanism

under 2 C cycle aging," *J. Power Sources*, vol. 196, no. 23, pp. 10336–10343, 2011.

[9] M. Dubarry *et al.*, "Identifying battery aging mechanisms in large format Li ion cells," *J. Power Sources*, vol. 196, no. 7, pp. 3420–3425, 2011.

[10] M. Dubarry, C. Truchot, and B. Y. Liaw, "Cell degradation in commercial LiFePO4cells with high-power and high-energy designs," *J. Power Sources*, vol. 258, pp. 408–419, 2014.

[11] A. Devie, M. Dubarry, and B. Y. Liaw, "Overcharge Study in Li4Ti5O12 Based Lithium-Ion Pouch Cell: I. Quantitative Diagnosis of Degradation Modes," *J. Electrochem. Soc.*, vol. 162, no. 6, pp. A1033–A1040, 2015.

[12] A. Devie and M. Dubarry, "Durability and reliability of electric vehicle batteries under electric utility grid operations. Part 1: Cell-to-cell variations and preliminary testing," *Batteries*, vol. 2, no. 3, pp. 1–13, 2016.

[13] M. Dubarry, A. Devie, and K. McKenzie, "Durability and reliability of electric vehicle batteries under electric utility grid operations: Bidirectional charging impact analysis," *J. Power Sources*, vol. 358, pp. 39–49, 2017.

[14] D. Anseán *et al.*, "Operando lithium plating quantification and early detection of a commercial LiFePO4cell cycled under dynamic driving schedule," *J. Power Sources*, 2017.

[15] A. Devie, G. Baure, and M. Dubarry, "Intrinsic variability in the degradation of a batch of commercial 18650 lithium-ion cells," *Energies*, vol. 11, no. 5, pp. 1–14, 2018.

[16] M. Dubarry, G. Baure, and A. Devie, "Durability and reliability of EV batteries under electric utility grid operations: Path dependence of battery degradation," *J. Electrochem. Soc.*, vol. 165, no. 5, pp. A773–A783, 2018.

[17] A. Barai *et al.*, "A comparison of methodologies for the non-invasive characterisation of commercial Li-ion cells," *Prog. Energy Combust. Sci.*, vol. 72, pp. 1–31, 2019.

[18] G. Baure and M. Dubarry, "Synthetic vs. Real driving cycles: A comparison of electric vehicle battery degradation," *Batteries*, vol. 5, no. 2, 2019.

[19] A. J. Smith, J. C. Burns, S. Trussler, and J. R. Dahn, "Precision Measurements of the Coulombic Efficiency of Lithium-Ion Batteries and of Electrode Materials for Lithium-Ion Batteries," *J. Electrochem. Soc.*, vol. 157, no. 2, p. A196, 2010.

[20] A. J. Smith and J. R. Dahn, "Delta Differential Capacity Analysis," *J. Electrochem. Soc.*, vol. 159, no. 3, pp. A290–A293, 2012.

[21] C. Weng, X. Feng, J. Sun, and H. Peng, "State-of-health monitoring of lithium-ion battery modules and packs via incremental capacity peak tracking," *Appl. Energy*, vol. 180, pp. 360–368, 2016.

[22] C. Weng, Y. Cui, J. Sun, and H. Peng, "On-board state of health monitoring of lithium-ion batteries using incremental capacity analysis with support vector regression," *J. Power Sources*, vol. 235, pp. 36–44, 2013.

[23] C. Pastor-Fernández, K. Uddin, G. H. Chouchelamane, W. D. Widanage, and J. Marco, "A Comparison between Electrochemical Impedance Spectroscopy and Incremental Capacity-Differential Voltage as Li-ion Diagnostic Techniques to Identify and Quantify the Effects of Degradation Modes within Battery Management Systems," *J. Power Sources*, vol. 360, pp. 301–318, 2017.

[24] M. Dubarry *et al.*, "Evaluation of commercial lithium-ion cells based on composite positive

electrode for plug-in hybrid electric vehicle applications. Part I: Initial characterizations," *J. Power Sources*, vol. 196, no. 23, pp. 10328–10335, 2011.

[25] Y. Merla, B. Wu, V. Yufit, N. P. Brandon, R. F. Martinez-Botas, and G. J. Offer, "Novel application of differential thermal voltammetry as an in-depth state-of-health diagnosis method for lithium-ion batteries," *J. Power Sources*, vol. 307, pp. 308–319, 2016.

[26] Y. Li *et al.*, "A quick on-line state of health estimation method for Li-ion battery with incremental capacity curves processed by Gaussian filter," *J. Power Sources*, 2018.

[27] E. Riviere, A. Sari, P. Venet, F. Meniere, and Y. Bultel, "Innovative incremental capacity analysis implementation for c/lifepo 4 cell state-of-health estimation in electrical vehicles," *Batteries*, vol. 5, no. 2, 2019.

[28] N. A. Samad, Y. Kim, J. B. Siegel, and A. G. Stefanopoulou, "Battery Capacity Fading Estimation Using a Force-Based Incremental Capacity Analysis," *J. Electrochem. Soc.*, vol. 163, no. 8, pp. A1584–A1594, 2016.

[29] S. Schindler, G. Baure, M. A. Danzer, and M. Dubarry, "Kinetics accommodation in Li-ion mechanistic modeling," *J. Power Sources*, vol. 440, no. August, p. 227117, 2019.

[30] I. Bloom *et al.*, "Differential voltage analyses of high-power, lithium-ion cells 1. Technique and application," *J. Power Sources*, vol. 139, no. 1–2, pp. 295–303, 2005.

[31] H. M. Dahn, A. J. Smith, J. C. Burns, D. A. Stevens, and J. R. Dahn, "User-Friendly Differential Voltage Analysis Freeware for the Analysis of Degradation Mechanisms in Li-Ion Batteries," *J. Electrochem. Soc.*, vol. 159, no. 9, pp. A1405–A1409, 2012.

[32] M. Lewerenz, A. Marongiu, A. Warnecke, and D. U. Sauer, "Differential voltage analysis as a tool for analyzing inhomogeneous aging: A case study for LiFePO4|Graphite cylindrical cells," *J. Power Sources*, vol. 368, pp. 57–67, 2017.

[33] M. Lewerenz, G. Fuchs, L. Becker, and D. U. Sauer, "Irreversible calendar aging and quantification of the reversible capacity loss caused by anode overhang," *J. Energy Storage*, vol. 18, no. April, pp. 149–159, 2018.

[34] M. Lewerenz and D. U. Sauer, "Evaluation of cyclic aging tests of prismatic automotive LiNiMnCoO2-Graphite cells considering influence of homogeneity and anode overhang," *J. Energy Storage*, vol. 18, no. June, pp. 421–434, 2018.

[35] B. Epding, B. Rumberg, H. Jahnke, I. Stradtmann, and A. Kwade, "Investigation of significant capacity recovery effects due to long rest periods during high current cyclic aging tests in automotive lithium ion cells and their influence on lifetime," *J. Energy Storage*, vol. 22, no. February, pp. 249–256, 2019.

[36] B. Gyenes, D. A. Stevens, V. L. Chevrier, and J. R. Dahn, "Understanding anomalous behavior in coulombic efficiency measurements on Li-ion batteries," *J. Electrochem. Soc.*, vol. 162, no. 3, pp. A278–A283, 2015.

[37] Mathworks, "MATLAB documentation: Findpeaks function," 2018. [Online]. Available: https://uk.mathworks.com/help/signal/ref/findpeaks.html.

[38] J. Remmlinger, M. Buchholz, M. Meiler, P. Bernreuter, and K. Dietmayer, "State-of-health monitoring of lithium-ion batteries in electric vehicles by on-board internal resistance estimation," *J. Power Sources*, vol. 196, no. 12, pp. 5357–5363, 2011.

[39] G. J. Offer, V. Yufit, D. A. Howey, B. Wu, and N. P. Brandon, "Module design and fault diagnosis in electric vehicle batteries," *J. Power Sources*, vol. 206, pp. 383–392, 2012.

[40] A. Jossen, “Fundamentals of battery dynamics,” *J. Power Sources*, vol. 154, no. 2, pp. 530–538, 2006.

[41] T.-J. Park, J.-B. Lim, and J.-T. Son, “Effect of Calcination Temperature of Size Controlled Microstructure of LiNi0.8Co0.15Al0.05O2 Cathode for Rechargeable Lithium Battery,” *Bull. Korean Chem. Soc.*, vol. 35, no. 2, p. 357, 2014.

[42] V. A. Sethuraman, L. J. Hardwick, V. Srinivasan, and R. Kostecki, “Surface structural disordering in graphite upon lithium intercalation/deintercalation,” *J. Power Sources*, vol. 195, no. 11, pp. 3655–3660, 2010.

[43] D. A. Howey, P. D. Mitcheson, V. Yufit, G. J. Offer, and N. P. Brandon, “Online Measurement of Battery Impedance Using Motor Controller Excitation,” *IEEE Trans. Veh. Technol.*, vol. 63, no. 6, pp. 2557–2566, Jul. 2014.

[44] Panasonic, “Datasheet : Specifications Powercell NCR18650B,” 2012.

[45] M. Lewerenz, P. Dechent, and D. U. Sauer, “Investigation of capacity recovery during rest period at different states-of-charge after cycle life test for prismatic Li(Ni 1/3 Mn 1/3 Co 1/3)O 2 -graphite cells,” *J. Energy Storage*, vol. 21, no. January, pp. 680–690, 2019.